

Section 6: Complete Analysis of BaseApp Base Class

1. Course Objectives

Before developing an App, we must understand the "father" of all Apps—`BaseApp`. It helps us handle most of the repetitive work. In this lesson, we will study the implementation of `base_app.py` in detail.

2. Core Source Code Analysis

File Path: `src/canmv/port/builtin_py/ybMain/base_app.py`

2.1 Initialization (`__init__`)

```
class BaseApp:
    def __init__(self, app_manager, name, icon=None):
        self.app_manager = app_manager # Save appmanager reference for
        convenient return to desktop later
        self.name = name                # App name
        self.icon = icon                 # App icon object
        self.screen = None               # App's main screen (created later)

        # Get screen resolution
        self.disp = lv.disp_get_default()
        self.disp_w = self.disp.get_hor_res()
        self.disp_h = self.disp.get_ver_res()
```

Note: In the `__init__` phase, we usually **don't create UI**. Because the system initializes all Apps at once during startup, if each App creates hundreds of controls here, memory might instantly explode (possibly).

2.2 Creating Desktop Icon (`create_icon`)

This method is called by `AppManager`.

```
def create_icon(self, parent):
    self.icon_btn = lv.btn(parent) # Create a button

    # Set icon text
    icon_label = lv.label(self.icon_btn)
    icon_label.set_text(self.name)

    # Bind click event
    self.icon_btn.add_event(self.on_icon_clicked, lv.EVENT.CLICKED, None)
    return self.icon_btn
```

2.3 Launch and Animation (1launch_with_animation)

This is the most exciting part of `BaseApp`. When you click the icon, it doesn't jump abruptly, but has an animation that zooms from the icon position to full screen.

```
def 1launch_with_animation(self):
    # 1. Create new screen (this is when the App's UI container is actually
    created)
    self.screen = lv.obj()

    # 2. Get current position of icon
    icon_coords = lv.area_t()
    self.icon_btn.get_coords(icon_coords)

    # 3. Set initial state: size equals icon, position equals icon
    self.screen.set_size(icon_w, icon_h)
    self.screen.set_pos(icon_x, icon_y)
    self.screen.set_style_radius(start_radius, 0) # Circular

    # 4. Create animation
    anim = lv.anim_t()
    # ... Set animation to zoom from icon_size to display_size ...

    # 5. After animation ends, call initialize()
    # This means subclasses must implement the initialize() method
```

3. What to Pay Attention to When Developing Apps?

After inheriting `BaseApp`, you only need to care about two things:

1. **Prepare Data:** Prepare the name and icon in `__init__`.
2. **Build Interface:** Override the `initialize(self)` method and draw your application interface on `self.screen`.